

Publications

Oluwaseun Olanike Ajayi

Current Affiliation: University of Georgia, Athens, Georgia, USA

Research Areas:

Bioanalytical Chemistry • Glycoproteomics • Protein Enzymology • Structural Biology •
Crystallography • Mass Spectrometry • Biopharmaceutical Innovation • Sustainable Analytical
Method Development

Peer-Reviewed Journal Publications 2025

Ajayi, O. O., & Phillips, R. S. (2025).

Investigating Cofactor Incorporation in AprL Aminotransferase: Redox, Chemical, and Cofactor Form Dependencies.

Free Radical Biology and Medicine.

Ajayi, O. O., & Ajayi, V. O. (2025).

A Systematic Optimization Analysis of N-Linked Glycans on Porous Graphitic Carbon (PGC) at Different Column Temperatures.

Journal of Progress in Engineering and Physical Science.

Aduloju, T. A., Ezekwesili-Ofili, J. O., Ajayi, O. O., & Oladejo, A. A. (2025).

Potential of Pumpkin (*Cucurbita maxima*) Fruit Used as Functional Food for the Management of Diabetes and Dyslipidemia.

European Journal of Nutrition & Food Safety.

Aduloju, T. A., Ezekwesili-Ofili, J. O., Oladejo, A. A., & Ajayi, O. O. (2025).

Modulation of Glycemic and Lipid Metabolic Pathways by *Cucurbita maxima* Fruit Parts in Streptozotocin-Induced Diabetic Rats.

Journal of Complementary and Alternative Medical Research.

Ajayi, O. O., & Ajayi, V. O. (2025).

Comparison of Porous Graphitic Carbon (PGC), Hydrophilic Interaction Liquid Chromatography (HILIC), and Reversed Phase (RP) Columns.

(Manuscript.)

Ajayi, V. O., Ameh, R. F., Sor, E. N., & Ajayi, O. O. (2025).
**Science Educators' Challenges of Fostering Students' Essential Skills for Global
Citizenship in Public Universities in Kogi State, Nigeria.**
SSRN Electronic Journal.

Ajayi, V. O., Ameh, R. F., & Ajayi, O. O. (2025).
**Application of Predict Explain Observe Explain Worksheet in Reconstructing Students'
Misconceptions in the Context of Alkanes and Alkanols.**
The Journal of Educational Paradigms.

Ajayi, V. O., Ameh, R. F., Chinasa, G. P., & Ajayi, O. O. (2025).
**Assessing Students' Misconceptions and HOTS Levels in the Context of Chemical Reaction
Using Paper-Based Five-Tier Test Diagnostic Instrument.**
Caribbean Journal of Education.

2024

Ajayi, O. O. (2024).
**A Review of Mass Spectrometry as an Important Analytical Tool for Structural
Elucidation of Biological Products.**
Journal of Progress in Engineering and Physical Science.

Book Chapters in Editorial Production

First author of all chapters. Invited book chapters currently under editorial review and/or print production with Springer Nature (Switzerland), CRC Press/Taylor & Francis (USA), and Routledge/CABI (United Kingdom).

1. Ajayi, O. O., Ajayi, V. O., Aisedion, M., & Penda, B. M.
**Chemical Industry: Biocatalysis and Enzyme Production, Synthesis of Fine Chemicals, and
Green Chemistry and Sustainable Processes.**
*In Prospects of Engineered Bacteria, Volume 2: Engineered Bacteria: Applications in Industry,
Chapter 6.*
Routledge/CAB International, United Kingdom.
Status: Submitted for publication; currently in editorial production.

2. Ajayi, O. O., Ajayi, V. O., & Aisedion, M.
**Processed Development of Microbial Cell Lines in Biopharmaceuticals: Innovations,
Challenges, and Future Directions.**

In Advances and Applications in Biotechnology, Volume 5: Biotechnology of Cell Lines in Biopharma, Chapter 2.

CRC Press/Taylor & Francis, USA.

Status: Submitted for publication; currently in editorial production.

3. Ajayi, O. O., Ajayi, V. O., & Aisedion, M.

An Alternative Strategy for the Use of Antimicrobials in Animals.

In Prospects of Fungal Biotechnologies for Livestock, Volume 4: Fungal Proteins and Enzyme Processes in Animal Husbandry, Chapter 9.

Springer Nature, Switzerland.

Status: Submitted for publication; currently under editorial review.

4. Ajayi, O. O., Ajayi, V. O., Salami, B. A., & Ayodeji, S. T.

Biochemical Reconstitution of AprL Aminotransferase: A Synthetic Biology Framework for PLP-Dependent Enzyme Engineering in Bacterial Systems.

In Prospects of Engineered Bacteria, Volume 1: Engineered Bacteria: Applications in Agriculture, Chapter 11.

Routledge/CAB International, United Kingdom.

Status: Submitted for publication; currently under editorial review.

5. Ajayi, O. O., Ajayi, V. O., & Aisedion, M.

Fungal Biotechnologies for Enhanced Protein Quality in Livestock Feed: Advances and Prospects.

In Prospects of Fungal Biotechnologies for Livestock, Volume 3: Myco-Engineered Food for Livestock, Chapter 11.

Springer Nature, Switzerland.

Status: Submitted for publication; currently under editorial review.

6. Ajayi, O. O., Ajayi, V. O., & Aisedion, M.

Enzyme-Responsive Delivery Platform: Focusing on Tumor-Specific Enzymes.

In Chemical Modulators of the Tumor Microenvironment: A Pharmaceutical Chemistry Perspective, Section 2: Pharmaceutical Chemistry Tools for Tumor Microenvironment Modulation, Topic 6.

Status: Submitted for publication; currently under editorial review.